

Test Bank NETWORK

CH1

1) What are the two most prominent types of packet switches in today's Internet?

- A) Modems and switches
- B) Hubs and bridges
- C) Routers and link-layer switches
- D) Servers and clients

Answer: C) Routers and link-layer switches

2) In the Internet protocol stack, application-layer protocols such as HTTP and SMTP are almost always implemented in software in the end systems.

- A) True
- B) False

Answer: A) True

3) Which of the following is NOT a type of delay experienced by packets in packet-switched networks?

- A) Processing delay
- B) Queuing delay
- C) Compilation delay
- D) Propagation delay

Answer: C) Compilation delay

4) Which access technology uses a combination of fiber optics and coaxial cable, often referred to as HFC?

- A) DSL
- B) FTTH
- C) Cable Internet access
- D) 5G Fixed Wireless

Answer: C) Cable Internet access

5) The propagation delay in a network is calculated as d/s , where d is the distance between routers and s is the propagation speed of the link.

- A) True
- B) False

Answer: A) True

6) Which type of multiplexing is used in circuit-switched networks?

- A) Packet multiplexing
- B) Frequency-division multiplexing (FDM) or time-division multiplexing (TDM)

- C) Space-division multiplexing
- D) Code-division multiplexing

Answer: B) Frequency-division multiplexing (FDM) or time-division multiplexing (TDM)

7) In the five-layer Internet protocol stack, which layers are typically implemented by link-layer switches?

- A) Layers 1 and 2
- B) Layers 1, 2, and 3
- C) All five layers
- D) Only layer 1

Answer: A) Layers 1 and 2

8) Consider a simple network with two end systems connected by a single router. If the source has three packets, each consisting of L bits, to send to the destination over links with transmission rate R , what is the total time for the destination to receive all three packets?

- A) $3L/R$
- B) $2L/R$
- C) $4L/R$
- D) $6L/R$

Answer: C) $4L/R$

9) In a network with N links between server and client, where each link has transmission rates $R_1, R_2, \dots, R_N$, what determines the throughput for a file transfer?

- A) The average of all transmission rates
- B) The sum of all transmission rates
- C) $\min\{R_1, R_2, \dots, R_N\}$, which is the transmission rate of the bottleneck link
- D) The maximum transmission rate among all links

Answer: C) $\min\{R_1, R_2, \dots, R_N\}$, which is the transmission rate of the bottleneck link

10) Which of the following statements about packet switching versus circuit switching is correct?

- A) Circuit switching offers better sharing of transmission capacity
- B) Packet switching is more suitable for voice calls due to predictable delays
- C) Packet switching offers better sharing of transmission capacity and is simpler, more efficient, and less costly to implement than circuit switching
- D) Circuit switching has variable end-to-end delays

Answer: C) Packet switching offers better sharing of transmission capacity and is simpler, more efficient, and less costly to implement than circuit switching

11) Tier-1 ISPs do not pay anyone as they are at the top of the hierarchy in the Internet's network structure.

- A) True
- B) False

Answer: A) True

12)What is encapsulation in the context of network protocols?

- A) Encrypting data for security
- B) Compressing data to reduce size
- C) The process where each layer adds its header information to the packet from the layer above, creating anew packet
- D) Removing unnecessary data from packets

Answer: C) The process where each layer adds its header information to the packet from the layer above, creating a new packet

13)Which physical medium is described as immune to electromagnetic interference, has very low signal attenuation up to 100 kilometers, and is very hard to tap?

- A) Twisted-pair copper wire
- B) Coaxial cable
- C) Fiber optics
- D) Terrestrial radio channels

Answer: C) Fiber optics

14)In Network Structure 5, which describes today's Internet, content-provider networks like Google attempt to bypass upper-tier ISPs by doing what?

- A) Building their own satellite networks
- B) Using only wireless connections
- C) Peering with lower-tier ISPs directly or at Internet Exchange Points (IXPs) and connecting to tier-1 ISPs for remaining access
- D) Eliminating the need for any ISP connections

Answer: C) Peering with lower-tier ISPs directly or at Internet Exchange Points (IXPs) and connecting totier-1 ISPs for remaining access

15)A distributed denial-of-service (DDoS) attack is harder to detect and defend against than a DoS attack from a single host.

- A) True
- B) False

Answer: A) True

16)Which of the following best describes the socket interface in Internet communication?

- A) A hardware component for network connections
- B) A physical port on end systems
- C) A set of rules that a sending program must follow so that the Internet can deliver data to a destination program on another end system
- D) A type of packet switch

Answer: C) A set of rules that a sending program must follow so that the Internet can deliver data to a destination program on another end system

17)End systems are also referred to as hosts because they host application programs such as Web browsers and email clients.

- A) True
- B) False

Answer: A) True

18)The Internet Engineering Task Force (IETF) develops Internet standards documented as RFCs(Requests for Comments).

- A) True
- B) False

Answer: A) True

19) What happens when the traffic intensity (λR) exceeds 1 in a packet-switched network?

- A) The network operates at optimal efficiency
- B) Packets are transmitted faster
- C) The queue will tend to increase without bound and queuing delay will approach infinity
- D) The network automatically reduces packet size

Answer: C) The queue will tend to increase without bound and queuing delay will approach infinity

20)DSL technology is expressly designed for short distances, and generally requires that residences be located within what distance of the central office (CO)?

- A) 1 to 2 miles
- B) 5 to 10 miles
- C) 15 to 20 miles
- D) 25 to 30 miles

Answer: B) 5 to 10 miles

CH2

21) socket is referred to as the.....between the application and the network

- A) isp
- B) Api
- C) dns
- D) vpn

Answer: B)Api

22) many multimedia applications are considered to be

- A) bandwidth-sensitive application
- B) real time-sensitive application
- C) elastic application
- D) jitter-intolerant application

Answer: A) bandwidth-sensitive application

23) TCP connection is a half-duplex connection

- A) true
- B) false

Answer: B) false

24) The TCP flow-control mechanism throttles a sending process (client or server) when the network is congested between sender and receiver

- A) true
- B) false

Answer: B) false

25) The default mode of HTTP uses

- A) persistent connections with pipelining
- B) persistent connections with non-parallel
- C) non-persistent with non-parallel
- D) non-persistent with parallel

Answer: A) persistent connections with pipelining

26) status codes 400 means ...

- A) The requested document does not exist on this server.
- B) The requested HTTP protocol version is not supported by the server.
- C) There is a generic error code indicating that the request could not be understood by the server.
- D) Request succeeded and the information is returned in the response

Answer: C) There is a generic error code indicating that the request could not be understood by the server.

27) A client sends the following HTTP request to the server:
GET /styles/main.css HTTP/1.1 Host: example.com If-Modified-Since: Tue, 02 Dec 2025 10:00:00 GMT

The file main.css on the server has a Last-Modified date of:

Tue, 01 Dec 2025 15:00:00 GMT

What status code should the server return?

- A) 304
- B) 200
- C) 505
- D) 301

Answer: A) 304

28) how can http/1.1 overcoming the Head of Line (HOL) blocking problem?

- A) by opening multiple parallel TCP connections
- B) by using framing mechanism
- C) by using congestion mechanism
- D) none of the above

Answer: A) by opening multiple parallel TCP connections

29) The primary goals for HTTP/2 ...

- A) HTTP/2 are to reduce perceived latency by enabling request and response multiplexing over a single TCP connection
- B) to get rid of (or at least reduce the number of) parallel TCP connections for transporting a single Web page
- C) using framing mechanism to avoid (HOL) blocking
- D) all of the above

Answer: D) all of the above

30) HTTP require multimedia data to be ASCII encoded before transfer.

- A) true
- B) false

Answer: B) false

31) The DNS protocol runs over ... and uses port

- A) TCP , 35
- B) TCP , 80
- C) UDP , 53
- D) UDP , 80

Answer: C) UDP , 53

32) In the Domain Name System (DNS) hierarchy, what is the role of a top-level domain (TLD) server?

- A) Providing the IP addresses of authorized DNS servers for a specific domain extension (e.g., .com or .edu).
- B) Providing the IP addresses of all root servers.
- C) Caching all recently resolved hostnames for the entire Internet Service Provider (ISP).
- D) Providing the final IP address for a specific hostname (e.g., www.google.com).

Answer: A) Providing the IP addresses of authorized DNS servers for a specific domain extension (e.g., .com or .edu).

33) Which type of DNS query involves a DNS server contacting other DNS servers on behalf of the client to resolve a domain name?

- A) Recursive
- B) Iterative
- C) Parallel
- D) Static

Answer: A) Recursive

34) A network entity that stores copies of recently requested objects in its local storage and satisfies HTTP requests on behalf of an origin server, often used by ISPs to reduce response time and traffic?

- A) cookies
- B) proxy server
- C) DNS server
- D) DNS caching

Answer: B) proxy server

35) UDP include a congestion-control mechanism

- A) true
- B) false

Answer: B) false

36) TCP is a no-frills, lightweight transport protocol

- A) true
- B) false

Answer: B) false

37) The RTT includes (multi choice)

- A) queuing delays
- B) processing delays
- C) propagation delays
- D) transmission delays

Answer: a ,b ,c

38) What is the primary function of cookies in the context of web browsing?

- A) Storing the entire webpage on the user's computer.
- B) Compressing webpage data for faster loading.
- C) Verifying user identity using encryption.
- D) Allowing the website to track user activity across multiple requests.

Answer: D) Allowing the website to track user activity across multiple requests.

39) What type of protocol is the Simple Mail Transfer Protocol (SMTP) primarily classified as?

- A) A "push" protocol, used to send email from the sender's mail server to the recipient's mail server.
- B) A peer-to-peer protocol for direct email exchange between users.
- C) A protocol for formatting the content of email messages.
- D) A "pull" protocol, used by a user agent to retrieve emails from the mailbox.

Answer: A) A "push" protocol, used to send email from the sender's mail server to the recipient's mail server.

40) Why does a user agent need a protocol like IMAP or HTTP in addition to SMTP to manage email?

- A) SMTP cannot handle attachments, so IMAP is used for files.
- B) SMTP is not secure, so HTTP with TLS is used instead.
- C) SMTP is for sending mail to the server, while IMAP or HTTP is needed to retrieve and manage mail from the server.
- D) IMAP is a newer and more efficient version of SMTP.

Answer: C) SMTP is for sending mail to the server, while IMAP or HTTP is needed to retrieve and manage mail from the server.

41) What is the main reason why peer-to-peer (P2P) file distribution architectures are considered "self-scaling"?

- A) Because every peer who joins to download a file also contributes their upload capacity to distribute the file to others.
- B) The file size is reduced for each peer that joins the network.
- C) All peers are required to have the same high-speed internet connection.
- D) The central server's capacity automatically increases as more peers join.

Answer: A) Because every peer who joins to download a file also contributes their upload capacity to distribute the file to others.

42)is the principal application-layer protocol for Internet electronic mail.

- A) FTP
- B) SMTP
- C) SIP
- D) Telnet

Answer: B) SMTP

Ch3

43) The transport layer provides logical communication between processes running on different hosts, while the network layer provides logical communication between hosts.

- A) True
- B) False

Answer: A) True

44) Which of the following statements about UDP and TCP is correct?

- A) UDP provides connection-oriented service while TCP provides connectionless service
- B) Both UDP and TCP provide reliable data transfer services to applications
- C) UDP provides unreliable, connectionless service while TCP provides reliable, connection-oriented service
- D) None of the above

Answer: C) UDP provides unreliable, connectionless service while TCP provides reliable, connection-oriented service

45) The process of delivering data in a transport-layer segment to the correct socket is called:

- A) Encapsulation
- B) Multiplexing
- C) Demultiplexing
- D) Segmentation

Answer: C) Demultiplexing

46) Well-known port numbers range from 0 to 1023 and are reserved for well-known application protocols such as HTTP and FTP.

- A) True
- B) False

Answer: A) True

47) In UDP, a socket is identified by which of the following?

- A) A two-tuple consisting of destination IP address and destination port number
- B) A four-tuple consisting of source IP address, source port number, destination IP address, and destination port number
- C) Only the destination port number
- D) A three-tuple consisting of source port number, destination IP address, and destination port number

Answer: A) A two-tuple consisting of destination IP address and destination port number

48) Which of the following is NOT an advantage of UDP over TCP?

- A) No connection establishment delay
- B) No connection state maintenance
- C) Guaranteed reliable data transfer
- D) Smaller packet header overhead

Answer: C) Guaranteed reliable data transfer

49) A TCP socket is identified by a four-tuple consisting of source IP address, source port number, destination IP address, and destination port number.

- A) True
- B) False

Answer: A) True

50) The UDP segment header consists of how many fields?

- A) Two fields
- B) Four fields
- C) Six fields
- D) Eight fields

Answer: B) Four fields

51) Why does UDP provide a checksum for error detection even though many link-layer protocols also provide error checking?

- A) Because UDP wants to be redundant
- B) Because there is no guarantee that all links between source and destination provide error checking, and bit errors could be introduced when a segment is stored in a router's memory
- C) Because link-layer protocols are unreliable
- D) None of the above

Answer: B) Because there is no guarantee that all links between source and destination provide error checking, and bit errors could be introduced when a segment is stored in a router's memory

51) In the rdt2.0 protocol, which type of acknowledgments are used?

- A) Only positive acknowledgments (ACK)
- B) Only negative acknowledgments (NAK)
- C) Both positive acknowledgments (ACK) and negative acknowledgments (NAK)
- D) Neither ACK nor NAK

Answer: C) Both positive acknowledgments (ACK) and negative acknowledgments (NAK)

52) The stop-and-wait protocol can have poor performance because the sender must wait for acknowledgment before sending the next packet.

- A) True
- B) False

Answer: A) True

53) In the rdt2.1 protocol, what is the purpose of adding sequence numbers to data packets?

- A) To ensure packets arrive in order
- B) To allow the receiver to determine whether a received packet is a retransmission or contains new data
- C) To implement flow control
- D) To detect corrupted packets

Answer: B) To allow the receiver to determine whether a received packet is a retransmission or contains new data

54) For a simple stop-and-wait protocol, a 1-bit sequence number is sufficient because:

- A) It's the smallest possible sequence number
- B) It allows the receiver to distinguish between a new packet and a retransmission in modulo-2 arithmetic
- C) It reduces header overhead
- D) All of the above

Answer: D) All of the above

55) In the rdt3.0 protocol (alternating-bit protocol), which mechanism is used to handle packet loss?

- A) Negative acknowledgments only
- B) Checksums only
- C) A timeout/retransmit mechanism where the sender retransmits if an ACK is not received within a timeout interval
- D) Forward error correction

Answer: C) A timeout/retransmit mechanism where the sender retransmits if an ACK is not received within a timeout interval

56) Pipelining techniques allow the sender to send multiple packets without waiting for acknowledgments, which significantly improves utilization compared to stop-and-wait protocols.

- A) True
- B) False

Answer: A) True

57) In a Go-Back-N (GBN) protocol, when a timeout occurs, the sender:

- A) Retransmits only the packet that timed out
- B) Retransmits all packets that have been previously sent but not yet acknowledged
- C) Sends a negative acknowledgment to the receiver
- D) Waits for another timeout before retransmitting

Answer: b)

58) In GBN protocol, the receiver discards out-of-order packets because:

- A) It simplifies receiver buffering since the receiver doesn't need to buffer any out-of-order packets
- B) Out-of-order packets are always corrupted
- C) The protocol specification requires it
- D) It improves network throughput

Answer: A) It simplifies receiver buffering since the receiver doesn't need to buffer any out-of-order packets

59) Which statement about Selective Repeat (SR) protocol is correct?

- A) SR retransmits all unacknowledged packets when a timeout occurs
- B) SR retransmits only those packets that it suspects were received in error, requiring the receiver to individually acknowledge correctly received packets
- C) SR does not use acknowledgments
- D) SR is identical to GBN

Answer: B) SR retransmits only those packets that it suspects were received in error, requiring the receiver to individually acknowledge correctly received packets

60) TCP provides full-duplex service, meaning application-layer data can flow from Process A to Process B at the same time as data flows from Process B to Process A.

- A) True
- B) False

Answer: A) True

61) The three-way handshake in TCP connection establishment involves:

- A) One segment from client to server
- B) Two segments exchanged between client and server
- C) Three segments exchanged where the client sends a SYN segment, the server responds with a SYNACK segment, and the client sends a final acknowledgment segment
- D) Four segments to ensure reliability

Answer: C) Three segments exchanged where the client sends a SYN segment, the server responds with a SYNACK segment, and the client sends a final acknowledgment segment

62) The Maximum Segment Size (MSS) in TCP refers to:

- A) The maximum size of the TCP segment including all headers
- B) The maximum amount of application-layer data in the segment, not including headers
- C) The maximum size of the TCP header
- D) The maximum size of the IP datagram

Answer: B) The maximum amount of application-layer data in the segment, not including headers

63) TCP provides cumulative acknowledgments, which means:

- A) TCP acknowledges every single packet individually
- B) TCP only acknowledges bytes up to the first missing byte in the stream
- C) TCP acknowledges only the last received packet

D) TCP doesn't use acknowledgments

Answer: B) TCP only acknowledges bytes up to the first missing byte in the stream

64) When a TCP sender receives three duplicate ACKs for the same data, it performs a fast retransmit by retransmitting the missing segment before that segment's timer expires.

- A) True
- B) False

Answer: A) True

65) The TCP timeout interval (Timeout Interval) is calculated as:

- A) EstimatedRTT only
- B) $\text{SampleRTT} \times 2$
- C) $\text{EstimatedRTT} + 4 \times \text{DevRTT}$
- D) $\text{DevRTT} \times 4$

Answer: C) $\text{EstimatedRTT} + 4 \times \text{DevRTT}$

67) TCP's error-recovery mechanism is best categorized as:

- A) Pure Go-Back-N protocol
- B) Pure Selective Repeat protocol
- C) A hybrid of GBN and SR protocols
- D) None of the above

Answer: C) A hybrid of GBN and SR protocols

68) Flow control in TCP is implemented to:

- A) Control congestion in the network
- B) Match the rate at which the sender is sending against the rate at which the receiving application is reading, preventing the sender from overflowing the receiver's buffer
- C) Ensure packets arrive in order
- D) Detect errors in transmitted segments

Answer: B) Match the rate at which the sender is sending against the rate at which the receiving application is reading, preventing the sender from overflowing the receiver's buffer

69) The receive window (rwnd) in TCP is used to give the sender an idea of how much free buffer space is available at the receiver.

- A) True
- B) False

Answer: A) True

70) When the TCP timeout interval expires and a segment is retransmitted, TCP sets the next timeout interval to:

- A) The same value as before

- B) Half the previous value
- C) Twice the previous value, causing intervals to grow exponentially after each retransmission
- D) A random value

Answer: C) Twice the previous value, causing intervals to grow exponentially after each retransmission

71) During TCP connection teardown, the TIME_WAIT state exists to:

- A) Allow the server to close properly
- B) Ensure all data has been transmitted
- C) Let the TCP client resend the final acknowledgment in case the ACK is lost
- D) Prevent new connections from starting

Answer: C) Let the TCP client resend the final acknowledgment in case the ACK is lost

72) When a host receives a TCP segment whose port numbers or source IP address do not match with any ongoing sockets in the host, the host:

- A) Silently discards the segment
- B) Sends an ICMP error message
- C) Sends a special reset segment with the RST flag bit set to 1 telling the source not to resend the segment
- D) Buffers the segment for future processing

Answer: C) Sends a special reset segment with the RST flag bit set to 1 telling the source not to resend the segment

Ch4

73) The network layer in H1 encapsulates transport-layer segments into datagrams before sending them to the router.

- A) True
- B) False

Answer: A

74) Routing is typically done in hardware because it must operate at nanosecond timescales.

- A) True
- B) False

Answer: B>false

75) A router uses its forwarding table by

- A) Examining packet header fields and using them to select the correct output interface
- B) Determining end-to-end routes between H1 and H2

- C) Encrypting and decrypting packet headers
- D) Checking for transport-layer errors before forwarding

Answer: A) Examining packet header fields and using them to select the correct output interface

76) In the traditional approach, each router runs a routing algorithm that communicates with other routers' routing algorithms.

- A) True
- B) False

Answer: A) True

77) In the SDN approach, which component is responsible for computing and distributing forwarding tables?

- A) Each individual router
- B) A centralized remote controller
- C) The transport-layer protocol
- D) DNS servers

Answer: B) A centralized remote controller

78) Which statement best describes a key difference between the traditional control plane and the SDN control plane?

- A) SDN removes the need for forwarding tables entirely.
- B) Traditional routing relies on a centralized controller instead of distributed routers.
- C) SDN separates routing logic from routers, relocating it to a software-based controller.
- D) Traditional routing algorithms run slower than SDN controllers.

Answer: C) SDN separates routing logic from routers, relocating it to a software-based controller.

79) Which router component is responsible for consulting the forwarding table to determine the correct output port?

- A) Routing processor
- B) Input port
- C) Switching fabric
- D) Output port

Answer: B) Input port

80) Which statement best describes the division of hardware vs. software responsibilities in a router?

- A) Input ports, output ports, and switching fabric are implemented in software; routing protocols run in hardware.
- B) Data-plane forwarding operates on millisecond timescales and is implemented in software.
- C) Control-plane functions run on the routing processor and operate at slower timescales compared to data-plane forwarding.
- D) The switching fabric is controlled entirely by a remote SDN controller.

Answer: C) Control-plane functions run on the routing processor and operate at slower timescales compared to data-plane forwarding.

81) The routing processor handles physical- and link-layer functions for incoming packets.

- A) True
- B) False

Answer: B)False

82) Why is a complete forwarding table with one entry for each possible 32-bit IP address impossible?

- A) It would require too much memory
- B) IP addresses do not use 32-bit values
- C) Forwarding tables cannot be stored on line cards
- D) The routing processor cannot compute them

Answer: A) It would require too much memory

83) Which of the following is NOT performed during input port processing?

- A) Checking and updating the packet's TTL field
- B) Performing physical- and link-layer processing
- C) Running the routing protocol to compute forwarding tables
- D) Updating network management counters

Answer: C) Running the routing protocol to compute forwarding tables

84) A router has the following forwarding table:

Prefix	Interface
1100	0
110010	1
11001000	2
(default)	3

**A packet arrives with destination address starting with:
1100100011010110...**

Which interface will the router forward the packet to?

- A) Interface 0
- B) Interface 1
- C) Interface 2
- D) Interface 3

Answer: C) Interface 2

85) In switching via a shared bus, multiple packets from different input ports can be transmitted simultaneously without waiting.

- A) True
- B) False

Answer: B) False

86) Which statement is true regarding switching via memory compared to switching via bus?

- A) Memory switching allows multiple packets to cross the bus simultaneously.
- B) Bus switching requires the routing processor to copy packets.
- C) Memory switching is limited by memory bandwidth, and only one packet can be read/written at a time.
- D) Bus switching can forward multiple packets in parallel without limitation.

Answer: C) Memory switching is limited by memory bandwidth, and only one packet can be read/written at a time.

87) Even if the switching fabric is N times faster than the input line speeds, output port queues can still form.

- A) True
- B) False

Answer: A) True

88) A router has 3 input ports and 1 output port. All input ports operate at the same line speed R_{line} , and the switch fabric operates at $3 \times R_{line}$. At a given time, one packet arrives at each input port, all destined for the single output port.

How many packets will be queued at the output port after the first time unit?

- A) 0
- B) 1
- C) 2
- D) 3

Answer: C) 2

89) Which of the following statements about buffering in routers is true?

- A) Larger buffers always reduce delay in the network.
- B) Buffering can absorb short-term traffic fluctuations but may increase queueing delay.
- C) Output port queues never experience packet loss if the switch fabric is fast enough.
- D) Head-of-the-line blocking occurs only at output queues.

Answer: B) Buffering can absorb short-term traffic fluctuations but may increase queueing delay.

90) In non-preemptive priority queuing, a higher-priority packet can interrupt the transmission of a currently transmitting lower-priority packet.

- A) True
- B) False

Answer: B) False

91) Consider a two-class round robin scheduler with Class 1 and Class 2 packets:

Packet arrivals (in order of time): 1 (Class 1), 2 (Class 1), 3 (Class 2), 4 (Class 1), 5 (Class 2)

Transmission takes 1 time unit per packet

In what order will packets be transmitted under round-robin scheduling?

- A) 1, 2, 3, 4, 5
- B) 1, 3, 2, 5, 4
- C) 1, 3, 2, 4, 5
- D) 1, 2, 3, 5, 4

Answer: C) 1, 3, 2, 4, 5

92) In CIDR notation, the "/24" in the address 223.1.1.0/24 indicates:

- A) The address has 24 host bits
- B) The network portion of the address is 24 bits
- C) There are 24 subnets
- D) There are 24 total addresses

Answer: B) The network portion of the address is 24 bits

93) The broadcast IP address 255.255.255.255 can be used to send a datagram to all hosts on a subnet.

- A) True
- B) False

Answer: B) True

94) When a DHCP server responds to a client's discover message, the server:

- A) Sends the response only to the client's MAC address
- B) Broadcasts the offer to all nodes on the subnet
- C) Sends the response directly to the next-hop router
- D) Waits for the client to request the IP before responding

Answer: B) Broadcasts the offer to all nodes on the subnet

95) Which of the following is NOT included in a DHCP offer message?

- A) Proposed IP address for the client
- B) Subnet mask
- C) Lease time
- D) MAC address of all other hosts

Answer: D) MAC address of all other hosts

96) When a NAT router forwards a datagram from an internal host to the Internet, it typically:

- A) Changes only the source IP address
- B) Changes only the destination IP address
- C) Changes both the source IP address and source port number
- D) Leaves the datagram unchanged

Answer: C) Changes both the source IP address and source port number

97) Which of the following is a main reason for developing IPv6?

- A) IPv4 addresses were running out
- B) To reduce the size of TCP headers
- C) To eliminate the need for routers
- D) To replace DNS

Answer: A) IPv4 addresses were running out

98) IPv6 routers perform fragmentation and reassembly of datagrams at intermediate routers, just like IPv4.

- A) True
- B) False

Answer: B) False

99) How does tunneling help in the IPv4-to-IPv6 transition?

- A) By allowing IPv6 datagrams to be sent inside IPv4 datagrams across IPv4 routers
- B) By converting IPv6 addresses to IPv4 addresses permanently
- C) By replacing all IPv4 routers with IPv6 routers instantly
- D) By compressing IPv6 headers to fit IPv4

Answer: A) By allowing IPv6 datagrams to be sent inside IPv4 datagrams across IPv4 routers

100) How does a NAT router know which internal host should receive a returning datagram from the Internet?

- A) By checking the destination IP address alone
- B) By checking the destination IP address and destination port number against its NAT translation table
- C) By using DNS lookup
- D) By sending the datagram to all hosts in the private network

Answer: B) By checking the destination IP address and destination port number against its NAT translation table

101) If a subnet does not have a DHCP server, what component helps the client communicate with a DHCP server on another subnet?

- A) Default gateway
- B) DHCP relay agent
- C) DNS server
- D) ARP cache

Answer: B) DHCP relay agent

102) Which of the following fields in the IPv6 header is used to identify a flow of datagrams for special handling, such as real-time services?

- A) Traffic class
- B) Flow label

- C) Hop limit
- D) Next header

Answer: B) Flow label

103) Which IPv6 header field is similar in purpose to the TTL field in IPv4?

- A) Traffic class
- B) Hop limit
- C) Flow label
- D) Next header

Answer: B) Hop limit

104) Which organization is responsible for the global allocation of IP addresses?

- A) IEEE
- B) ICANN
- C) IETF
- D) ISO

Answer: B) ICANN

105) A NAT-enabled router allows multiple devices on a private network to share a single public IP address.

- A) True
- B) False

Answer: A) True

106) Increasing buffer size at a router always reduces packet loss without affecting delay.

- A) True
- B) False

Answer: B) False

107) In priority queuing, what happens to packets with lower priority when high-priority packets continuously arrive?

- A) They are transmitted first
- B) They may experience starvation
- C) They are dropped immediately
- D) They increase the link bandwidth

Answer: B) They may experience starvation